

3. Magnesium alloys are used in the casings of tablet computers and mobile phones.

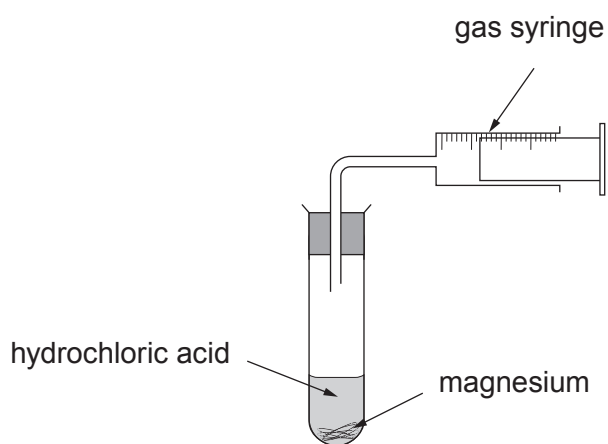
- (a) Complete the sentence below by underlining the correct word in the brackets. [2]

An alloy is a (**mixture** / **compound** / **solution**) of two or more elements. One of these must be a (**gas** / **metal** / **liquid**).

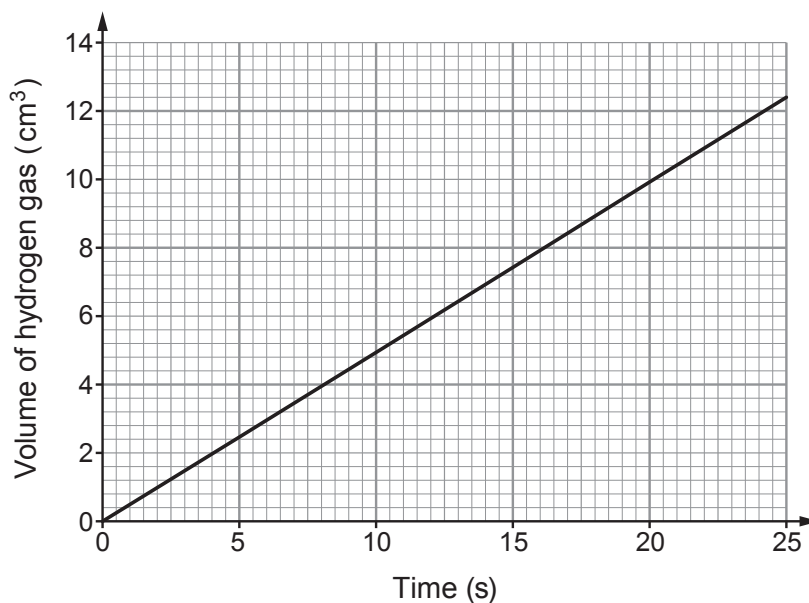
- (b) Magnesium reacts with hydrochloric acid producing hydrogen gas.



A student investigated this reaction using the apparatus below.



A graph of the student's results is shown below.



Draw a line on the graph above to show the results that you would expect if the experiment was repeated with a lower concentration of hydrochloric acid. [1]



- (c) The student also investigated the reaction of hydrochloric acid and magnesium at different initial temperatures.

Initial temperature (°C)	Time taken for magnesium to disappear (s)
10	80
20	40
30	20
40	10
50	5

- (i) State **three** variables that need to be controlled in this experiment. [3]

1.
2.
3.

- (ii) I. The student thinks that every time the temperature doubles, the time taken halves. Explain whether you agree. [2]

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- II. Predict the time it would take for the magnesium to disappear at 60 °C. [1]

time = s

- (iii) Complete the following sentences by underlining the correct words in the brackets. [2]

As the temperature increases, particles move
(**faster** / **slower** / **at the same speed**).

The particles collide (**more often** / **as often** / **less often**) in a given time.

